Held-Out SFT Clarification: Ten-Example Generation Sanity Check

Status: Completed by the owner; ten-example routing diagnostic, not a benchmark.

Why This Check

The name-first model produced no valid clarification decisions on the established 150-case development suite. A read-only audit found **zero raw outputs mentioning request_clarification**, compared with 26 raw mentions and 23 accepted clarification decisions for E. This is not simply a parser hiding malformed clarification calls.

Before changing training data, test whether this checkpoint can generate clarification on held-out inputs from the same SFT distribution. This does not retrain a model or alter the benchmark, prompt policy, parser, class counts or existing result files.

Selection and Inference

- Select the first 10 unique clarification prompts in the unchanged first 200 validation rows. The selection is deterministic, not a representative random sample.
- Validate the supplied target/tool contracts and system-prompt hash.
- Reject exact input overlap with the first 1,000 raw training rows actually used for this experiment. This is not a semantic/paraphrase leakage audit.
- Pass only the saved system/user messages, unchanged, to inference. The expected assistant answer is recorded for inspection but is never supplied to the model.
- Reuse the same LocalLoraAgent loading, chat-template, generation and strict parser path as benchmark evaluation: greedy decoding, 256 new tokens, no output repair.
- Check adapter metadata against the selected config and record config, dataset, validation-file and adapter hashes. No tool calls are executed.
- Record raw output, parsed decision and expected target separately for every row. Count clarification decisions, not wording keywords or full task success.

Commands

Run from the repository root in PowerShell:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --output tmp\name-first-heldout-clarification-10.json
```

The default config is the trained two-epoch QKVO name-first candidate. Its expected adapter path comes from that config. No extra adapter path or base-model override is needed. CUDA is required unless explicitly allowing slow CPU inference.

For selection/preflight checks without loading a model or tokenizer:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --dry-run `
  --output tmp\name-first-heldout-clarification-setup.json
```

Use a new output filename for subsequent runs; existing diagnostics are not overwritten. Reports stay under ignored `tmp/` because they contain local SFT inputs and expected targets, and are not published benchmark artifacts.

Observed Result

The saved owner-run report is `tmp/name-first-heldout-clarification-10.json`:

Recorded Prediction	Examples
CLARIFY	7
ACT	3
REFUSE	0
Invalid/unknown	0

All ten expected CLARIFY. The same seven raw outputs mentioned `request_clarification`. This establishes that the checkpoint can generate valid clarification calls on these selected held-out inputs. It does not establish that the seven questions are adequate or that clarification transfers to the benchmark.

Unblinded inspection found clear question-quality gaps: row 25 asks only for the start time when duration is also missing; row 42 asks for the task name/status instead of the missing due date, despite the title being supplied. These are qualitative observations, not an independent semantic review or revised scores.

Interpretation

If clarification is generated here but absent from benchmark outputs, investigate transfer and differences in the model-visible input distribution. If clarification is also absent here, investigate the checkpoint and learning/generation behavior before assuming the problem is only transfer to the benchmark.

Neither outcome establishes the cause by itself. Ten selected examples cannot prove general competence or inability. Valid clarification syntax is not evidence that the question is sufficient or appropriate; inspect the saved raw outputs and expected targets. Adapter metadata/hash checks identify the requested checkpoint, but do not constitute an independent numerical comparison of all loaded weights.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.